

Informative Power-System Probing via Sequential Bayesian Experimental Design

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Abstract—Stable power-system operation benefits from reduced uncertainty in inertia and frequency-response parameters. Active probing helps identify these parameters, but its information yield depends on the inputs. We apply sequential Bayesian optimal experimental design to adapt probe durations to observed responses and obtain more information with the same number of probes. On IEEE9 and IEEE14 reduced models, we compare offline-trained adaptive policies (DAD and RL-sBOED), online policy refinement (Step-DAD), one-step Bayesian design (Myopic), an optimized fixed sequence, and random probing. With three to five probes, the learned methods achieve higher mean EIG than the three baselines. DAD and RL-sBOED also require substantially less online computation than Myopic and Step-DAD. Online refinement gives modest additional gains on IEEE9 but no consistent improvement on IEEE14, where estimator saturation limits method discrimination. The results support offline-trained sequential design for informative probing with low online computation.

Index Terms—Bayesian experimental design, power-system identification, adaptive probing, expected information gain, rate of change of frequency.

I. INTRODUCTION

Stable power-system operation supports reliable electricity supply. Engineers need information about inertia M and effective frequency response K to predict disturbance responses and choose operating and control actions. Reducing uncertainty in these parameters can support better informed decisions. Active probing identifies them by injecting controlled signals and measuring frequency responses [1].

The information obtained depends on the probing inputs. Our aim is to learn more about M and K with the same number of experiments by adapting successive probes to observed responses. Sequential Bayesian optimal experimental design (sBOED) provides the framework. Fixed designs cannot use new observations; myopic designs use them but optimize only the next experiment [2]. Full-sequence design can account for how each probe informs later choices. It must also account for the grid's physical state changes caused by preceding probes.

Existing methods offer different information–computation trade-offs. Deep adaptive design (DAD) learns a policy offline to maximize information over a complete sequence [3]. RL-sBOED learns policies through simulated rewards [4]; Step-DAD refines a pretrained policy after observations arrive [5]. We evaluate whether these methods yield more information

under matched grid probing constraints, and whether online refinement justifies its cost.

We formulate continuous probe-duration selection with a shared sensor model and physical state carried between experiments. Six existing methods are compared on reduced IEEE9 and IEEE14 systems at equal probe budgets. We report EIG, seed variability, and offline and online computation. Our contribution is the power-system design formulation and matched evaluation; the policy algorithms are existing methods, and conclusions are limited to the stated models and sensor.

II. PROBLEM FORMULATION

We seek to reduce uncertainty in M and K through informative probes within a budget of T design steps. Here $t = 1, \dots, T$ indexes design steps, d denotes elapsed time within each probe window, and D_t is the selected pulse duration.

A. Unknown Parameters and Power-System Model

For a reduced network with m machines, the unknown parameter vector is $\theta = (M_1, \dots, M_m, K_1, \dots, K_m)$, where M_i and K_i are inertia and effective frequency-response coefficients. The prior $p_0(\theta)$ has independent components $M_i \sim \mathcal{U}(0.7\bar{M}_i, 1.3\bar{M}_i)$ and $K_i \sim \mathcal{U}(0.1\bar{K}_i, \bar{K}_i)$. One draw of θ remains fixed throughout all T probes.

Let δ_i be rotor angle, ω_i angular-speed deviation, and u_i the applied probing power. The reduced swing model is

$$\begin{aligned} \dot{\delta}_i &= \omega_i, \\ M_i \dot{\omega}_i &= u_i - \left(D_i + \frac{K_i}{2\pi} \right) \omega_i \\ &\quad - \sum_{j=1}^m B_{ij} \sin(\delta_i - \delta_j). \end{aligned} \tag{1}$$

Dots denote derivatives with respect to physical time d . The damping D_i and network couplings B_{ij} are known; frequency deviation is $\Delta f_i = \omega_i/(2\pi)$ in Hz. The units of M_i , K_i and D_i are pu s²/rad, pu/Hz and pu s/rad, respectively, on a 100-MVA base at nominal frequency 60 Hz. All angles and speed deviations start at zero. Net mechanical injections are zero in this deviation model, making the initial unforced state an equilibrium.

TABLE I
PRIOR SCALES AND DAMPING COEFFICIENTS

Network	Bus	$\bar{M}_i$	$\bar{K}_i$	D_i
IEEE9	1	0.1254141	0.2937500	0.0125414
	2	0.0339531	0.1312500	0.0033953
	3	0.0159685	0.0750000	0.0015969
IEEE14	1	0.0212207	0.0784314	0.0021221
	2	0.0344836	0.1274510	0.0034484
	3, 6, 8	0.0265257	0.0980392	0.0026526

IEEE9 retains buses 1, 2 and 3; IEEE14 retains buses 1, 2, 3, 6 and 8. Test-case reactances [6] define a lossless Laplacian $\mathbf{L}$. Eliminating algebraic buses l while retaining machines g gives $\mathbf{L}_{\text{red}} = \mathbf{L}_{gg} - \mathbf{L}_{gl}\mathbf{L}_{ll}^{-1}\mathbf{L}_{lg}$, with $B_{ij} = -(\mathbf{L}_{\text{red}})_{ij}$ for $i \neq j$ and $B_{ii} = 0$. Resistance, voltage dynamics and transformer tap effects are omitted. Table I summarizes the model parameterization. Nominal inertia is computed as $\bar{M}_i = 2H_i/(2\pi f_0)$, with $f_0 = 60$ Hz. For IEEE9, $H = (23.64, 6.40, 3.01)$ s matches published machine data [7, Table 2]. For IEEE14, $H = (4, 6.5, 5, 5, 5)$ s is prescribed at buses (1, 2, 3, 6, 8) as a benchmark setting, without a verified literature source for these values. For both networks, $\bar{K}_i = K_{\text{tot}}w_i$, with $K_{\text{tot}} = 0.5$ pu/Hz. The allocation weights are $w = (0.5875, 0.2625, 0.15)$ for IEEE9 and $w = (8, 13, 10, 10, 10)/51$ for IEEE14. Fixed damping satisfies $D_i = (0.1 \text{ s}^{-1})\bar{M}_i$. The response scales, damping and prior ranges are shared study assumptions, not validated operating-grid parameters. Table entries are rounded; configurations retain full precision. Training and evaluation draw independently from the same prior.

B. Probe Design and Observation Model

The probe design is $\xi_t = (A_t, B_t, D_t)$: amplitude, injection bus and pulse duration. This study fixes $A_t = 0.05$ pu and $B_t = 1$; only D_t is optimized. The raised-cosine input for $0 \leq d \leq D_t$ is

$$u_{B_t,t}(d) = \frac{A_t}{2} [1 - \cos(2\pi d/D_t)]. \quad (2)$$

The input is zero for $D_t < d \leq D_{\text{obs}}$ and at all other buses. The observation window is $D_{\text{obs}} = 3.5$ s. The full angle and speed state at its end becomes the initial state of the next window, without resetting the system. Consequently, the response depends on the complete design sequence $\xi_{1:t} = (\xi_1, \dots, \xi_t)$.

Write $\Delta f_{1,t}(d; \theta, \xi_{1:t})$ for the resulting bus-1 frequency deviation. The noise-free response is the signed endpoint rate of change of frequency (RoCoF), denoted by y_t :

$$y_t(\theta, \xi_{1:t}) = \frac{\Delta f_{1,t}(D_{\text{obs}}) - \Delta f_{1,t}(D_{\text{obs}} - \Delta d_{\text{samp}})}{\Delta d_{\text{samp}}}. \quad (3)$$

The frequency arguments $\theta, \xi_{1:t}$ are suppressed in the numerator. With sampling interval $\Delta d_{\text{samp}} = 0.025$ s, the measured response follows

$$y_{\text{obs},t} = y_t(\theta, \xi_{1:t}) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2), \quad (4)$$

where $\sigma = 0.005$ Hz/s and noise is independent across windows. Each observation becomes available at window end. Decision latency is idealized as zero in simulated time.

C. Sequential Decisions and Bayesian Updating

The observation model connects the physical experiment to inference and subsequent decisions. The data after step t are the ordered design–observation history $h_t = ((\xi_k, y_{\text{obs},k}))_{k=1}^t$, with $h_0 = \emptyset$. A policy π selects ξ_t using only observed history h_{t-1} , with amplitude and bus fixed as above. It has no access to true parameters or physical state. Feasible durations satisfy

$$\begin{aligned} 0.2 &\leq D_t \leq 3, \quad t = 1, \dots, T, \\ D_t - D_{t-1} &\geq \Delta D_{\min}, \quad t = 2, \dots, T, \end{aligned} \quad (5)$$

where $\Delta D_{\min} = 0.01$ s. At step t , the upper limit is tightened to $3 - (T - t)\Delta D_{\min}$ to leave room for the remaining probes; the lower limit is 0.2 initially and $D_{t-1} + \Delta D_{\min}$ thereafter. All methods map a unit action into this interval without sorting or discretizing durations. Let Π denote the policies satisfying these constraints; any policy randomness is independent of θ .

The observation model defines a Gaussian likelihood $p(y_{\text{obs},t} | \theta, h_{t-1}, \xi_t)$ with mean $y_t(\theta, \xi_{1:t})$ and variance σ^2 . Bayes' rule updates parameter uncertainty after each measurement:

$$p(\theta | h_t) = \frac{p(y_{\text{obs},t} | \theta, h_{t-1}, \xi_t) p(\theta | h_{t-1})}{p(y_{\text{obs},t} | h_{t-1}, \xi_t)}. \quad (6)$$

The denominator is the predictive density, obtained by integrating the numerator over θ , and $p(\theta | h_0) = p_0(\theta)$. Each candidate parameter vector propagates its own physical state under the observed designs to evaluate the likelihood. Conditional on the history, design selection provides no additional knowledge of θ ; policy probabilities therefore cancel from the Bayesian update.

D. Expected Information Gain and Design Objective

To compare possible design policies, we quantify the uncertainty reduction produced by their observations using the Kullback–Leibler divergence of the terminal posterior from the prior. The expected information gain (EIG) of policy π is

$$\mathcal{I}_T(\pi) = \mathbb{E}_{h_T|\pi} [D_{\text{KL}}(p(\theta | h_T) \| p_0(\theta))]. \quad (7)$$

Here $D_{\text{KL}}(p\|q) = \int p(\theta) \log[p(\theta)/q(\theta)] d\theta$; natural logarithms express information in nats. The expectation averages over parameter draws, measurement noise and any policy randomness. The sequential Bayesian design problem is

$$\pi^* \in \arg \max_{\pi \in \Pi} \mathcal{I}_T(\pi). \quad (8)$$

Thus all methods receive the same number of probes and seek greater expected reduction in parameter uncertainty. Offline and online computation are measured separately from this information objective.

Exact EIG requires intractable marginal likelihoods. We use the sequential prior-contrastive (sPCE) lower bound [3] for training and evaluation. Let $\theta^{(0)} \sim p_0$ generate the observed history and let $\theta^{(1)}, \dots, \theta^{(L)}$ be independent prior contrasts.

TABLE II
COMPARISON OF EXPERIMENTAL DESIGN METHODS

Method	Adaptive	Myopic	Optimization
Random	No	N/A	None
Fixed [2]	No	No	Offline
Myopic [2]	Yes	Yes	Online
DAD [3]	Yes	No	Offline
RL-sBOED [4]	Yes	No	Offline
Step-DAD [5]	Yes	No	Both

For any candidate θ , define the observation likelihood along the realized sequence as

$$\mathcal{L}_t(\theta) = \prod_{k=1}^t p(y_{\text{obs},k} \mid \theta, h_{k-1}, \xi_k). \quad (9)$$

The sampled information score is then

$$\hat{I}_t = \log \frac{\mathcal{L}_t(\theta^{(0)})}{\frac{1}{L+1} \sum_{\ell=0}^L \mathcal{L}_t(\theta^{(\ell)})}. \quad (10)$$

Likelihoods are evaluated in the log domain. Averaging $\hat{I}_T$ over simulated sequences estimates a lower bound on $\mathcal{I}_T(\pi)$; For IEEE9 and IEEE14, reported EIG values are estimated using this sPCE lower bound; they are not exact EIG values. Individual scores can be negative but cannot exceed $\log(L+1)$. With $L = 128$, this ceiling is 4.8598 nats, so near-ceiling scores can conceal differences in true EIG. The generating parameter is used only by the evaluator, never supplied to an online planner.

III. COMPARED DESIGN METHODS

The objective in (8) motivates three comparisons: optimized versus random probing, adaptive versus fixed designs, and offline policy learning versus online optimization. The six existing methods below select $\xi_t = (0.05, 1, D_t)$ using the same feasible duration interval (5) and observation model. Table II distinguishes adaptation to observations from myopic optimization of only the next probe. Fixed optimizes the full sequence without observation feedback. Random optimizes no information objective, so its myopia entry is N/A.

A. Nonadaptive Designs: Random and Fixed

Random draws $z_t \sim \mathcal{U}(0, 1)$ independently at each step and sets $D_t = D_t^{\min} + (D_t^{\max} - D_t^{\min})z_t$. Here $D_1^{\min} = 0.2$, $D_t^{\min} = D_{t-1} + \Delta D_{\min}$ for $t > 1$, and $D_t^{\max} = 3 - (T - t)\Delta D_{\min}$. Thus sampling is uniform within the remaining feasible interval, not over complete ordered sequences. It ignores observations and requires no offline training.

Fixed is a static Bayesian design: all durations are optimized jointly before observations are collected [2, Sec. 2.4]. Our implementation optimizes the ordered sequence using EIG for the full sequence. It compares short, central, long and spread interior initializations using validation, then optimizes the selected initialization. Each training seed yields a separately fitted sequence. During evaluation, Fixed executes that sequence without adapting to observations.

B. Online One-Step Design: Myopic

Unlike Fixed, Myopic uses observation feedback. It follows one-step adaptive BOED, maximizing the expected information from the next observation under the current posterior [2, Sec. 2.4]. Our implementation maintains a weighted particle approximation to this posterior. Before each decision after the first, particles propagate their own states under the preceding duration and update likelihood weights using the latest observation. Continuous search then maximizes expected information from the next probe only, using 16 proposals, three search rounds and 32 simulated observations per proposal. Candidate durations obey the same constraints as all other methods. No offline policy is trained.

C. Offline-Trained Adaptive Policies: DAD and RL-sBOED

DAD and RL-sBOED retain observation feedback while optimizing over the full sequence offline. Both policies receive the stage index and ordered slots containing past durations, scaled observations and availability masks; future slots are masked. They do not receive true parameters or physical states. At evaluation, a forward pass produces the action without online optimization or an explicit particle-posterior update.

DAD [3] uses a deterministic two-layer, 64-unit tanh network, a trainable stage bias and a sigmoid action map. Offline training maximizes EIG by differentiating through the complete simulated sequence, including observations and carried states. In the joint Fixed/DAD campaign, validation selects between random and interior Fixed-based initialization. The reported DAD stage timer excludes the preceding Fixed training; the full workflow incurs both costs.

RL-sBOED [4] uses a REDQ-style actor-critic adaptation with ten critics and five replay updates per trajectory batch. The actor has two 128-unit ReLU layers. Its per-step information rewards are $R_t = \hat{I}_t - \hat{I}_{t-1}$, with $\hat{I}_0 = 0$ and discount $\gamma = 1$, so their sum estimates full-sequence EIG. Training therefore accounts for future design steps without differentiating through the simulator. Entropy regularization uses normalized tanh actions and target entropy -1 ; reported information scores exclude entropy bonuses. Evaluation uses the actor’s deterministic output.

D. Online Policy Refinement: Step-DAD

Step-DAD [5] tests whether adapting policy weights to the observed history improves on offline training alone. Each episode begins with the matching selected DAD checkpoint and uses it unchanged for the first probe. Before each subsequent decision, it assimilates the latest observation into its 128-particle posterior. It then samples candidate parameters and their carried states from that posterior to simulate the remaining design steps. These future simulations condition on the actual observed history.

At each refinement, a fresh Adam optimizer performs four gradient updates with batch size 32 and learning rate 10^{-3} . A fixed conditional validation batch compares the unchanged policy and all updated candidates; the best is retained. Retained weights carry to the next design step within the episode, but

not to another episode. The total refinement budget is $4(T - 1)$ updates: 8, 12 and 16 for $T = 3, 4, 5$. The planner never receives the evaluator’s true parameters.

For both Myopic and Step-DAD, assimilation is performed only when a subsequent decision requires it. No final planner update is needed after $y_{\text{obs},T}$ for this EIG experiment; terminal information is computed separately by the evaluator using (10).

IV. EXPERIMENTAL SETTINGS

To assess these design strategies at equal measurement budgets, we compare all six methods at $T = 3, 4, 5$ design steps on IEEE9 and IEEE14. Each step applies one probe and collects one observation. The systems share sensor and design constraints but use separate physical parameters and trained policies.

Figure 1 connects probing to inference: observations reweight candidate parameters through their likelihoods, and adaptive designs use the resulting history to select later probes.

The swing equations are integrated by RK4 with step $1/640$ s. Numerical state/duration Jacobians use perturbations of 10^{-5} and include carried-state derivatives. The verification suite compares policy and conditional-refinement gradients with end-to-end finite differences.

Offline training uses 2,000 updates with batch size 32 and three independent training seeds (101, 202 and 303). The DAD and Fixed learning rate is 10^{-3} . Validation uses 128 systems every 100 updates. Training, validation and evaluation use 128 contrastive samples for the sPCE objective. Myopic and Step-DAD use 128 particles for online posterior planning; Step-DAD performs four refinement updates after each observation when design steps remain. Evaluation uses seeds 1001, 1002 and 1003, with 512 simulated experimental sequences per evaluation seed. Validation selects checkpoints without gradient updates on validation cases. Each trained policy is evaluated on 1,536 episodes. Parameter and measurement-noise draws are paired across methods; the online planner uses a separate random stream. Random/Myopic have no training-seed dependence under these streams, so identical evaluations can be shared across training seeds. Fixed remains independently optimized for each training seed.

For trained methods, we average evaluation-seed means within each training seed and report mean $\pm$ sample standard deviation across three trainings. Random/Myopic report mean $\pm$ sample SD of evaluation-seed means, not episode-level dispersion. Evaluation seed 1001 informed development and is not untouched test data.

Experiments use NVIDIA RTX 5080 and A100 GPUs, with RTX 4090 used for development. IEEE9 offline times combine RTX 5080 and A100 runs; online timings use RTX 5080 except Step-DAD $T = 5$, training seed 303, which ran on A100. IEEE14 timings use A100. These recorded workflow costs do not provide a controlled runtime comparison between networks. Configurations, source hashes and checkpoints preserve provenance. Code: https://github.com/alexschmidt123/tpc2027_power_grid.

V. RESULTS AND DISCUSSION

We first compare information acquired at equal probe budgets, then examine the computation required to obtain it. Optimization diagnostics and estimator limitations qualify the resulting comparisons.

A. Information Versus Number of Design Steps

Tables III and IV report EIG in nats as mean $\pm$ sample SD. Trained methods use three training-seed means; Random and Myopic use three evaluation-seed means. Both networks use $L = 128$ contrasts, giving a 4.8598-nat estimator ceiling.

TABLE III
IEEE9 EIG

Method	$T = 3$	$T = 4$	$T = 5$
Random	0.4321 ± 0.0310	0.4924 ± 0.0342	0.5392 ± 0.0471
Fixed	2.0112 ± 0.0116	2.3192 ± 0.0004	2.5066 ± 0.0013
Myopic	1.9736 ± 0.0438	2.1707 ± 0.0366	2.2407 ± 0.0547
DAD	2.1366 ± 0.0161	2.4095 ± 0.0080	2.6351 ± 0.0179
RL-sBOED	2.1257 ± 0.0027	2.3963 ± 0.0333	2.5307 ± 0.0114
Step-DAD	2.1471 ± 0.0110	2.4382 ± 0.0048	2.6621 ± 0.0145

TABLE IV
IEEE14 EIG

Method	$T = 3$	$T = 4$	$T = 5$
Random	0.7541 ± 0.0047	0.8360 ± 0.0297	0.8464 ± 0.0252
Fixed	4.2623 ± 0.0402	4.4946 ± 0.0345	4.6624 ± 0.0083
Myopic	4.2907 ± 0.0368	4.3990 ± 0.0177	4.4641 ± 0.0280
DAD	4.7576 ± 0.0362	4.8325 ± 0.0047	4.8443 ± 0.0026
RL-sBOED	4.7445 ± 0.0022	4.8275 ± 0.0112	4.8383 ± 0.0108
Step-DAD	4.7527 ± 0.0275	4.8297 ± 0.0067	4.8446 ± 0.0042

On IEEE9, DAD exceeds Fixed by 0.1253, 0.0903 and 0.1286 nats at $T = 3, 4, 5$, respectively. The observed Step-DAD gains over DAD are 0.0105, 0.0287 and 0.0269 nats at $T = 3, 4, 5$, respectively. These are modest mean gains; their computational cost is examined below. IEEE9 scores remain well below the estimator ceiling.

For IEEE14, Table IV shows higher mean EIG for DAD, RL-sBOED and Step-DAD than for Random, Fixed or Myopic at every design-step count. At $T = 3$, DAD exceeds Myopic by 0.4669 nats and Fixed by 0.4953 nats. The comparison fixes the number of probes, so the gain comes from selecting more informative experiments within the same probe budget. Fixed exceeds Myopic at $T = 4, 5$; observation adaptation alone therefore does not guarantee a stronger design.

The DAD-family means are close to one another. Step-DAD changes the DAD mean by -0.0049 , -0.0028 and $+0.0003$ nats for $T = 3, 4, 5$, respectively. These small differences do not establish an advantage for online refinement. At $T = 5$, DAD lies only 0.0155 nats below the 4.8598-nat estimator ceiling. Finite-contrast saturation can compress method differences; these scores neither measure exact EIG nor establish a

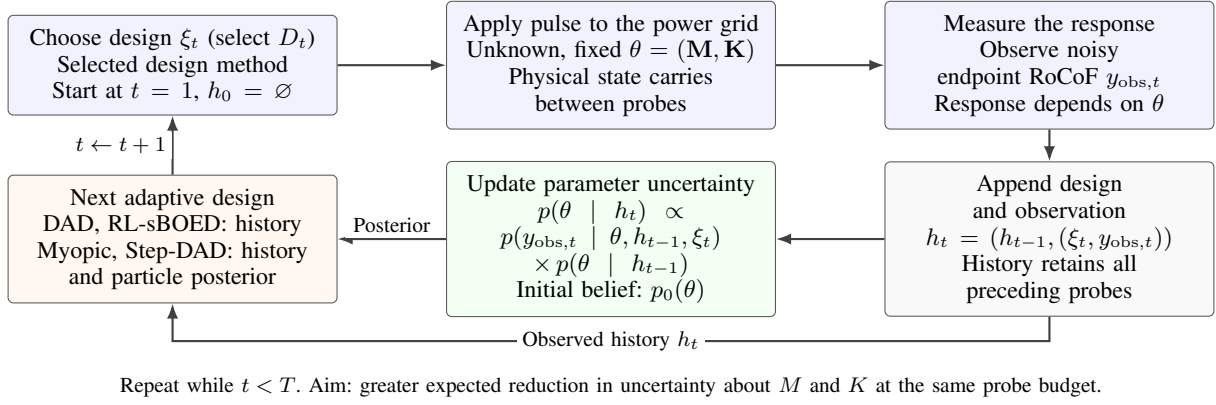


Fig. 1. Sequential probing and information feedback. Each response extends the history, which determines the posterior through the likelihood. The next adaptive design uses this history directly (DAD/RL-sBOED), or the history and its particle posterior for online planning (Myopic/Step-DAD). DAD/RL-sBOED do not explicitly compute a posterior. Random and Fixed use the same physical experiment but bypass adaptive feedback. Stop after T observations; EIG is estimated separately, without requiring a final planner update. Uncertainty reduction is expected over experiments, not guaranteed for every observation.

reliable ranking within the DAD family. The SDs describe distinct sources of variation: retraining for learned/fitted methods and repeated evaluation for Random/Myopic.

B. Measured Computational Cost

Higher information scores are useful only in relation to the resources required to select the probes. We therefore separate the cost of learning a reusable policy from the cost incurred during each experiment. Tables V and VI report offline training in hours; Tables VII and VIII report online computation in milliseconds per complete sequence. Online cost is the wall-clock time summed over all T decisions in one experimental sequence. For each evaluation seed, we average 512 episodes per trained policy and then average the three training seeds where applicable. Tables report the mean and sample SD across the three evaluation-seed means; this SD describes evaluation variability, not retraining variability or episode-level dispersion. Online cost includes action selection, posterior updates needed for subsequent decisions, Myopic search and Step-DAD refinement. It excludes the simulated physical response, observation acquisition and EIG estimation. These are computational timings, not end-to-end physical experiment durations.

TABLE V
IEEE9 OFFLINE TRAINING TIME

Method	$T = 3$	$T = 4$	$T = 5$
Random	0	0	0
Fixed	0.960 ± 0.352	1.396 ± 0.515	1.827 ± 0.662
Myopic	0	0	0
DAD	0.957 ± 0.342	1.393 ± 0.509	1.831 ± 0.654
RL-sBOED	0.120 ± 0.017	0.152 ± 0.030	0.180 ± 0.045
Step-DAD	0.957 ± 0.342	1.393 ± 0.509	1.831 ± 0.654

The offline timer includes initialization selection, optimization, periodic validation and checkpoint writes. DAD's reported stage time excludes the separately incurred Fixed

TABLE VI
IEEE14 OFFLINE TRAINING TIME

Method	$T = 3$	$T = 4$	$T = 5$
Random	0	0	0
Fixed	1.877 ± 0.011	2.750 ± 0.002	3.691 ± 0.050
Myopic	0	0	0
DAD	1.894 ± 0.032	2.725 ± 0.018	3.626 ± 0.068
RL-sBOED	0.170 ± 0.008	0.204 ± 0.001	0.248 ± 0.001
Step-DAD	1.894 ± 0.032	2.725 ± 0.018	3.626 ± 0.068

TABLE VII
IEEE9 ONLINE COMPUTATION TIME

Method	$T = 3$	$T = 4$	$T = 5$
Random	0.0715 ± 0.0001	0.09445 ± 0.00003	0.121 ± 0.001
Fixed	0.523 ± 0.038	0.623 ± 0.002	0.766 ± 0.002
Myopic	9308 ± 4	12577 ± 2	15892 ± 2
DAD	0.537 ± 0.002	0.683 ± 0.002	0.838 ± 0.004
RL-sBOED	0.492 ± 0.003	0.626 ± 0.001	0.764 ± 0.002
Step-DAD	6342 ± 2	15808 ± 9	24350 ± 50

TABLE VIII
IEEE14 ONLINE COMPUTATION TIME

Method	$T = 3$	$T = 4$	$T = 5$
Random	0.161 ± 0.004	0.209 ± 0.002	0.309 ± 0.049
Fixed	1.16 ± 0.14	1.48 ± 0.04	2.75 ± 1.73
Myopic	10000 ± 2	13437 ± 9	16935 ± 35
DAD	1.22 ± 0.09	1.42 ± 0.02	1.93 ± 0.20
RL-sBOED	1.68 ± 0.29	3.31 ± 1.33	2.71 ± 0.69
Step-DAD	9113 ± 7	23243 ± 24	44693 ± 32

training used for initialization; a joint Fixed-DAD workflow must include both costs. Step-DAD reuses the matching DAD model: its offline entries repeat the DAD training cost, with no additional training run; refinement costs are online. Offline means and sample SDs use three training seeds. Random and

Myopic require no offline training, so their entries are zero.

At $T = 5$, DAD requires 1.93 ms online per sequence, compared with 16.94 s for Myopic and 44.69 s for Step-DAD (Table VIII). Thus the near-identical DAD/Step-DAD scores have markedly different computational costs. Offline training amortizes design computation over repeated experiments, but this benefit depends on how often a trained policy is reused and on the physical deadline. The current simulation assumes zero decision latency; it does not demonstrate real-time feasibility for the online planners.

C. Optimization and Estimator Diagnostics

The information–computation comparison also depends on policy optimization and the numerical information estimator. Evaluation uses the best validation checkpoint; histories are retained in the result collection. DAD’s update count excludes Fixed initialization training, whose cost is reported separately.

For IEEE14 training seed 101, the DAD and RL-sBOED checkpoints occur at updates 1,200 and 1,000, respectively, whereas Fixed selects update 2,000. Later updates need not improve validation utility. Neither a fixed update budget nor the recorded validation histories establish convergence. Contrast-count and planner-particle sensitivity remain to be tested; being below the estimator ceiling does not establish estimator accuracy.

D. Scope and Limitations

Beyond these numerical considerations, the reduced lossless network, independent priors and scalar sensor restrict the conclusions to the stated simulation model. Parameter distributions match between training and evaluation; robustness to prior shift, sensor bias and unmodeled dynamics is not established by this protocol. Equal update budgets imply neither equal convergence nor equal computation.

Finite-contrast scoring and finite-particle inference introduce approximation errors distinct from ODE integration error. Information gain establishes neither terminal-control safety nor field performance.

VI. CONCLUSION

Under the specified IEEE9 and IEEE14 models, DAD, RL-sBOED and Step-DAD achieve higher mean EIG than Random, Fixed and Myopic at equal budgets of three to five probes. Offline-trained DAD and RL-sBOED also require low online computation. Four-update Step-DAD gives modest mean gains over DAD on IEEE9, but no consistent improvement on IEEE14, while adding substantial online cost.

These findings support adaptive probing to reduce parameter uncertainty as a step toward better informed operation; improved stability or control performance is not evaluated here. Near-ceiling IEEE14 scores prevent strong rankings within the DAD family, and development use of evaluation seed 1001 limits independent validation. Larger contrast counts, model mismatch and decision latency are priorities for further evaluation. Optional MoE-sBOED development remains outside this six-method comparison.

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APPENDIX

SIR-ODE BENCHMARK

These historical results are retained for reference only; they do not support the power-grid conclusions. SIR-ODE uses a population of 500, two initial infections, 100 candidate measurement times, observation noise SD 1.0, and a precomputed simulation bank. The objective remains EIG, estimated here by averaging prior-to-posterior entropy reductions over a 1,024-particle representation. This estimator has ceiling $\log(1024) = 6.9315$ nats; the grid experiments instead use sPCE.

Table IX reports EIG as mean $\pm$ sample SD in nats. DAD/RL-sBOED/Step-DAD use three training-seed means; Random/Myopic/Fixed use five evaluation-seed means after deduplication. Each evaluation uses 512 systems; Random averages 32 design replicates per system. Fixed is calibrated once. Historical training uses a behavioral-cloning warm start, different optimizers, and checkpoint selection with a sequence-diversity preference. Full settings, models and results remain in the companion collection.

TABLE IX
SIR-ODE EIG

Method	$T = 3$	$T = 4$	$T = 5$
Random	3.7783 ± 0.0167	3.8593 ± 0.0167	3.9241 ± 0.0167
Fixed	6.2816 ± 0.0101	6.4501 ± 0.0091	6.5360 ± 0.0044
Myopic	6.7108 ± 0.0125	6.7437 ± 0.0124	6.7596 ± 0.0135
DAD	6.8394 ± 0.0087	6.8591 ± 0.0039	6.8658 ± 0.0077
RL-sBOED	6.8494 ± 0.0029	6.8674 ± 0.0020	6.8814 ± 0.0015
Step-DAD	6.8438 ± 0.0095	6.8679 ± 0.0011	6.8761 ± 0.0109